

T3 matched EFT Lagrangian terms: EFT_1_after_F_S1_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F, S1, S2. Remaining active T3 fields: none.

Field combinations 1 of 5

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{\hbar H_{a_3} H_{a_4} H_{a_1}^\dagger H_{a_2}^\dagger \mathcal{I}_{H1,a_1,a_3}^1 \mathcal{I}_{H1,a_2,a_4}^1 \left(\lambda_{H1}^{(1)}\right)^2}{2\epsilon}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon}$
								$+\frac{3\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(2)}\right)^2}{2\epsilon}$		
cont.								$+\frac{\text{CH21} H_{a_1} H_{a_1}^\dagger - H_{a_1} \mu^2 H_{a_1}^\dagger}{2\epsilon}$		
								$+\frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(1)}\right)^2}{2\epsilon}$	$-\frac{5\hbar B_{\mu_1\mu_2}^2}{12\epsilon}$	$-\frac{\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}$
								$+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(2)}\right)^2}{6\epsilon}$		
cont.								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(1)}\right)^2}{2\epsilon}$		
								$+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(2)}\right)^2}{6\epsilon}$		
cont.								$+\frac{C_{H2} H_{a_1} H_{a_1}^\dagger - H_{a_1} \mu^2 H_{a_1}^\dagger}{2\epsilon}$		
								$+\frac{4\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon}$		
cont.								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \lambda_{H1}^{(1)} \lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$		
								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \lambda_{H2}^{(1)} \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$		
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(1)}\right)^2}{2\epsilon}$	$-\frac{11\hbar B_{\mu_1\mu_2}^2}{12\epsilon}$	$-\frac{25\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}$
								$+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(2)}\right)^2}{6\epsilon}$		
cont.								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(1)}\right)^2}{2\epsilon}$		
								$+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(2)}\right)^2}{6\epsilon}$		
cont.								$+\frac{C_{H2} H_{a_1} H_{a_1}^\dagger - H_{a_1} \mu^2 H_{a_1}^\dagger}{2\epsilon}$		
								$+\frac{4\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon}$		
cont.								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \lambda_{H1}^{(1)} \lambda_{H1}^{(2)}}{\sqrt{3}\epsilon}$		
								$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \lambda_{H2}^{(1)} \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{3\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(2)}\right)^2}{2\epsilon}$	$-\frac{7\hbar B_{\mu_1\mu_2}^2}{12\epsilon}$	$-\frac{2\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon}$
								$+\frac{\hbar H_{a_3} H_{a_4} H_{a_1}^\dagger H_{a_2}^\dagger \mathcal{I}_{H2,a_1,a_3} \mathcal{I}_{H2,a_2,a_4} \left(\lambda_{H2}^{(1)}\right)^2}{2\epsilon}$		
cont.								$+\frac{+C_{H2} H_{a_1} H_{a_1}^\dagger - H_{a_1} \mu^2 H_{a_1}^\dagger}{\sqrt{3}\epsilon}$		
								$+\frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$		
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}^{(2)}\right)^2}{2\epsilon}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{5\hbar \left(W_{\mu_1\mu_2}^{A_1}\right)^2}{6\epsilon}$
								$+\frac{3\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}^{(2)}\right)^2}{2\epsilon}$		
cont.								$+\frac{+C_{H2} H_{a_1} H_{a_1}^\dagger - H_{a_1} \mu^2 H_{a_1}^\dagger}{\epsilon}$		
								$+\frac{2\hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$		

Field combinations 2 of 5

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 H$	$S_1 W_{\mu\nu}$	$S_2 H$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H1,a_1,a_3}{}^1\mathcal{I}_{H1,a_2,a_4}{}^1\left(\lambda_{H1}^{(1)}\right)^2$		$+\frac{3}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}^{(2)}\right)^2$
								$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}^{(1)}\right)^2$		$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}^{(1)}\right)^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{5}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}^{(2)}\right)^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{5}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}^{(2)}\right)^2$
								$\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}^{(1)}\lambda_{H1}^{(2)}$		$\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}^{(1)}\lambda_{H2}^{(2)}$
								$+\frac{\sqrt{3}}{\sqrt{3}}$		$+\frac{\sqrt{3}}{\sqrt{3}}$
								$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}^{(1)}\right)^2$		$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}^{(1)}\right)^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{5}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}^{(2)}\right)^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{5}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}^{(2)}\right)^2$
								$\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}^{(1)}\lambda_{H1}^{(2)}$		$\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}^{(1)}\lambda_{H2}^{(2)}$
								$+\frac{\sqrt{3}}{\sqrt{3}}$		$+\frac{\sqrt{3}}{\sqrt{3}}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{3}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}^{(2)}\right)^2$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H2,a_1,a_3}{}^1\mathcal{I}_{H2,a_2,a_4}{}^1\left(\lambda_{H2}^{(1)}\right)^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}^{(2)}\right)^2$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{3}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}^{(2)}\right)^2$

Field combinations 3 of 5

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 B_{\mu\nu}$	$S_2 W_{\mu\nu}$	$F B_{\mu\nu}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{24}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{3}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{12}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$		$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$

Field combinations 4 of 5

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FW_{\mu\nu}$	$H \ell e$	$S_1 S_2 H$
cont.									$+\frac{\sqrt{3}hy_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	
									$-\frac{1}{8}\sqrt{3}hy_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}hy_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\left(M_{S_1}^2-M_{S_2}^2\right)}$
									$-\frac{1}{8}\sqrt{3}hy_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$+ \frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\left(M_{S_1}^2-M_{S_2}^2\right)}$
cont.									$+\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\left(M_{S_1}^2-M_{S_2}^2\right)}$
									$-\frac{1}{8}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$- \frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\left(M_{S_1}^2-M_{S_2}^2\right)}$
cont.									$+\frac{\sqrt{3}hy_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	
									$-\frac{1}{8}\sqrt{3}hy_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	
cont.									$+\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	
									$-\frac{1}{8}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}hy_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$
									$-\frac{1}{8}\sqrt{3}hy_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$+ \frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$
cont.									$+\frac{\sqrt{3}hy_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$
									$-\frac{1}{8}\sqrt{3}hy_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$- \frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$
cont.									$+\frac{\sqrt{3}hy_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	
									$-\frac{1}{8}\sqrt{3}hy_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$	
cont.									$+\frac{\sqrt{3}hy_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$	
									$-\frac{1}{8}\sqrt{3}hy_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$	

Field combinations 5 of 5

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	S_1	F	H	ℓ	S_1	F	H	ℓ	S_2	F	H	ℓ
T3-A	0	1	0	3	1	2	$\frac{1}{2}$												
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S2}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$- \frac{2h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S1}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \lambda \tau_3 L_{a_3,1,1} C P_R \overline{L}_{a_4,1,2} e^T M_{S2}^2 \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} e_{a_1,1,3} e_{a_2,1,4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
cont.								$- \frac{2h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \lambda \tau_3 L_{a_3,1,1} C P_R \overline{L}_{a_4,1,2} e^T M_{S1}^2 \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} e_{a_1,1,3} e_{a_2,1,4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S1}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$- \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S1}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \lambda \tau_3 L_{a_3,1,1} C P_R \overline{L}_{a_4,1,2} e^T M_{S1}^2 M_{S2}^2 \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} e_{a_1,1,3} e_{a_2,1,4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
cont.								$- \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \lambda \tau_3 L_{a_3,1,1} C P_R \overline{L}_{a_4,1,2} e^T M_{S1}^2 M_{S2}^2 \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} e_{a_1,1,3} e_{a_2,1,4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S1}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$- \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S1}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \lambda \tau_3 L_{a_3,1,1} C P_R \overline{L}_{a_4,1,2} e^T M_{S1}^2 M_{S2}^2 \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} e_{a_1,1,3} e_{a_2,1,4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
cont.								$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{Y}_{e,1,3,1,2} \bar{e}_{1,2} P_L L_{a_1,1,1} \overline{y}_{1,1,3,1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$+ \frac{h \overline{y}_{1,1,1,\tau_1} \overline{Y}_{e,1,3,1,2} \bar{e}_{1,2} P_L L_{a_1,1,1} M_{S1}^2 \overline{y}_{1,1,3,1} H_{a_1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$+ \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{Y}_{e,1,3,1,2} \bar{e}_{1,2} P_L L_{a_1,1,1} M_{S1}^2 \overline{y}_{1,1,3,1} H_{a_1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$- \frac{h \overline{y}_{1,1,1,\tau_1} \overline{Y}_{e,1,3,1,2} \bar{e}_{1,2} P_L L_{a_1,1,1} M_{S1}^2 \overline{y}_{1,1,3,1} H_{a_1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
cont.								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_{S1}^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_{S1}^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
cont.								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_{S1}^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_{S1}^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
cont.								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_{S1}^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_{S1}^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
								$- \frac{h \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{2 \left(M_F^2 - M_{S1}^2 \right)}$											
T3-B	0	2	0	2	1	1	$\frac{1}{2}$												
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S2}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$- \frac{2h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S1}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \lambda \tau_3 L_{a_3,1,1} C P_R \overline{L}_{a_4,1,2} e^T M_{S2}^2 \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} e_{a_1,1,3} e_{a_2,1,4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
cont.								$- \frac{2h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \lambda \tau_3 L_{a_3,1,1} C P_R \overline{L}_{a_4,1,2} e^T M_{S1}^2 M_{S2}^2 \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} e_{a_1,1,3} e_{a_2,1,4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S1}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$- \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} L_{a_4,1,2} e^T C P_L L_{a_2,1,1} H_{a_1} H_{a_3} M_{S1}^2 \lambda \tau_3 e_{a_1,1,2} e_{a_3,1,4} M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
								$+ \frac{2h \log \left(\frac{M_F^2}{M_{S2}^2} \right) \lambda \tau_3 L_{a_3,1,1} C P_R \overline{L}_{a_4,1,2} e^T M_{S1}^2 M_{S2}^2 \overline{y}_{1,1,1,\tau_1} \overline{y}_{2,1,2,\tau_1} e_{a_1,1,3} e_{a_2,1,4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\sqrt{3} \left(M_F^2 - M_{S1}^2 \right) \left(M_F^2 - M_{S2}^2 \right) \left(M_{S1}^2 - M_{S2}^2 \right)}$											
cont.								$- \frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
cont.								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											
								$- \frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y}_{1,1,3,1} \overline{L}_{a_1,1,2} P_R e_{1,1} H_{a_1} M_{S1}^2 \overline{y}_{1,1,2,1} \overline{Y}_{e,1,3,1}}{4 \left(M_F^2 - M_{S1}^2 \right)^2}$											

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F H \ell$	$S_1 F H \ell$	$S_2 F H \ell$	
cont.								$-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} L_{a3, r1}} C P_R \overline{L_{a4, i2}} M_{S_1}^2 M_{S_2}^2 y_{1, i1, r1} y_{2, i2, r1} \epsilon_{a1, a3} \epsilon_{a2, a4} H_{a1}^\dagger H_{a2}^\dagger M_F^2}{\sqrt{3} \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $+\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F}{\sqrt{3} \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F}{\sqrt{3} \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$	$+\frac{\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $+\frac{\hbar y_{1, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_1}^2 y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $+\frac{\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} M_{S_1}^2 y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $+\frac{\hbar y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} y_{1, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)^2}$	$+\frac{\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{\hbar y_{2, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_2}^2 y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} M_{S_2}^2 y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\hbar y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} y_{2, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$	
	cont.								$+\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} L_{a3, i1}} C P_R \overline{L_{a4, i2}} M_{S_1}^2 M_{S_2}^2 y_{1, i1, r1} y_{2, i2, r1} \epsilon_{a1, a3} \epsilon_{a2, a4} H_{a1}^\dagger H_{a2}^\dagger M_F}{\sqrt{3} \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} L_{a3, i1}} C P_R \overline{L_{a4, i2}} M_{S_1}^2 M_{S_2}^2 y_{1, i1, r1} y_{2, i2, r1} \epsilon_{a1, a3} \epsilon_{a2, a4} H_{a1}^\dagger H_{a2}^\dagger M_F}{\sqrt{3} \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$	$+\frac{\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} y_{1, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $+\frac{\hbar y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_1}^2 y_{1, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $+\frac{\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} M_{S_1}^2 y_{1, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $-\frac{\hbar y_{1, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_1}^2 y_{1, i2, r1} Y_{e, i3, i1}}{2 \left(M_F^2 - M_{S_1}^2 \right)}$	$+\frac{\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} y_{2, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{\hbar y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_2}^2 y_{2, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} M_{S_2}^2 y_{2, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{\hbar y_{2, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_2}^2 y_{2, i2, r1} Y_{e, i3, i1}}{2 \left(M_F^2 - M_{S_2}^2 \right)}$
	cont.								$-\frac{\hbar \log \left(\frac{\mu^2}{M_{S_1}^2} \right) \overline{y_{1, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} M_{S_1}^2 y_{1, i2, r1} Y_{e, i3, i1}}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $-\frac{\hbar y_{1, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_1}^4 y_{1, i2, r1} Y_{e, i3, i1}}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $-\frac{\hbar y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_1}^2 y_{1, i3, r1} H_{a1}^\dagger}{2 \left(M_F^2 - M_{S_1}^2 \right)}$	$-\frac{\hbar \log \left(\frac{\mu^2}{M_{S_2}^2} \right) \overline{y_{2, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} M_{S_2}^2 y_{2, i2, r1} Y_{e, i3, i1}}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $-\frac{\hbar y_{2, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_2}^4 y_{2, i2, r1} Y_{e, i3, i1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{\hbar y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_2}^2 y_{2, i3, r1} H_{a1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)}$	
cont.								$-\frac{\hbar \log \left(\frac{\mu^2}{M_{S_1}^2} \right) \overline{y_{1, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} M_{S_1}^4 y_{1, i2, r1} Y_{e, i3, i1}}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $-\frac{\hbar y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_1}^2 y_{1, i3, r1} H_{a1}^\dagger}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $-\frac{\hbar \log \left(\frac{\mu^2}{M_{S_1}^2} \right) \overline{y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} M_{S_1}^2 y_{1, i3, r1} H_{a1}^\dagger}{2 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $-\frac{\hbar y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_1}^4 y_{1, i3, r1} H_{a1}^\dagger}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$	$-\frac{\hbar \log \left(\frac{\mu^2}{M_{S_2}^2} \right) \overline{y_{2, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} M_{S_2}^4 y_{2, i2, r1} Y_{e, i3, i1}}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{\hbar y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_2}^2 y_{2, i3, r1} H_{a1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $-\frac{\hbar \log \left(\frac{\mu^2}{M_{S_2}^2} \right) \overline{y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} M_{S_2}^2 y_{2, i3, r1} H_{a1}^\dagger}{2 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $-\frac{\hbar y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_2}^4 y_{2, i3, r1} H_{a1}^\dagger}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$		
cont.								$-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F^2}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $+\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F^2}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} L_{a3, i1}} C P_R \overline{L_{a4, i2}} M_{S_1}^2 M_{S_2}^2 y_{1, i1, r1} y_{2, i2, r1} \epsilon_{a1, a3} \epsilon_{a2, a4} H_{a1}^\dagger H_{a2}^\dagger M_F^2}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $+\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} L_{a3, i1}} C P_R \overline{L_{a4, i2}} M_{S_1}^2 M_{S_2}^2 y_{1, i1, r1} y_{2, i2, r1} \epsilon_{a1, a3} \epsilon_{a2, a4} H_{a1}^\dagger H_{a2}^\dagger M_F^2}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $+\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$	$-\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar y_{1, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $+\frac{\sqrt{3}\hbar y_{1, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_1}^2 y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} M_{S_1}^2 y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $+\frac{\sqrt{3}\hbar y_{1, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_1}^4 y_{1, i2, r1} Y_{e, i3, i1} M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$	$-\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar y_{2, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{\sqrt{3}\hbar y_{2, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_2}^2 y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i3, r1} L_{a1, i2}} P_R e_{i1} H_{a1} M_{S_2}^2 y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{\sqrt{3}\hbar y_{2, i3, r1} L_{a1, i2} P_R e_{i1} H_{a1} M_{S_2}^4 y_{2, i2, r1} Y_{e, i3, i1} M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$	
cont.								$-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} L_{a3, i1}} C P_R \overline{L_{a4, i2}} M_{S_1}^2 M_{S_2}^2 y_{1, i1, r1} y_{2, i2, r1} \epsilon_{a1, a3} \epsilon_{a2, a4} H_{a1}^\dagger H_{a2}^\dagger M_F}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $+\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} L_{a3, i1}} C P_R \overline{L_{a4, i2}} M_{S_1}^2 M_{S_2}^2 y_{1, i1, r1} y_{2, i2, r1} \epsilon_{a1, a3} \epsilon_{a2, a4} H_{a1}^\dagger H_{a2}^\dagger M_F}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $+\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} L_{a3, i1}} C P_R \overline{L_{a4, i2}} M_{S_1}^2 M_{S_2}^2 y_{1, i1, r1} y_{2, i2, r1} \epsilon_{a1, a3} \epsilon_{a2, a4} H_{a1}^\dagger H_{a2}^\dagger M_F}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $-\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$ $+\frac{2\hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1, i1, r1} y_{2, i2, r1} L_{a4, i2}}{}^T C P_L L_{a2, i1} H_{a1} H_{a3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \epsilon_{a1, a2} \epsilon_{a3, a4} M_F}{3 \left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_2}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)}$	$+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} y_{1, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $+\frac{\sqrt{3}\hbar y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_1}^2 y_{1, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} M_{S_1}^2 y_{1, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_1}^2 \right)}$ $+\frac{\sqrt{3}\hbar y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_1}^4 y_{1, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} M_{S_1}^4 y_{1, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar y_{1, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_1}^6 y_{1, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_1}^2 \right)^2}$	$+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} y_{2, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{\sqrt{3}\hbar y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_2}^2 y_{2, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} M_{S_2}^2 y_{2, i3, r1} H_{a1}^\dagger M_F^2}{2 \left(M_F^2 - M_{S_2}^2 \right)}$ $+\frac{\sqrt{3}\hbar y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_2}^4 y_{2, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}}} P_L L_{a1, i1} M_{S_2}^4 y_{2, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$ $+\frac{\sqrt{3}\hbar y_{2, i1, r1} Y_{e, i3, i2} \overline{e_{i2}} P_L L_{a1, i1} M_{S_2}^6 y_{2, i3, r1} H_{a1}^\dagger M_F^2}{4 \left(M_F^2 - M_{S_2}^2 \right)^2}$	

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